COB vs MiP vs SMD CHEAT SHEET

LED display technology comparison for specifiers | BRIGHTLUXX

Specification	SMD	MiP	COB
Technology	Surface Mount Device	MicroLED in Package	Chip on Board
Contrast Ratio	3,000-5,000:1	8,000-12,000:1	15,000-20,000:1
Surface	Individual LED bumps	Smooth, low-profile	Fully smooth, matte
Pixel Pitch Range	P1.2 - P10+	P0.7 - P2.5	P0.4 - P1.9
Repairability	Single LED replace	Single LED replace	Module-level swap
Durability	Moderate (exposed LED)	High (protected)	Highest (fully encapsulated)
Power Consumption	Baseline	-20% vs SMD	-40% vs SMD
Lifespan	50,000-80,000 hrs	80,000-100,000 hrs	100,000+ hrs
Cost (relative)	$	$$	$$$
Best Application	Budget, large format	Mid-premium, versatile	Premium, close-view

WHEN TO SPECIFY EACH

SMD: Pixel pitch P2.5+, budget-sensitive projects, outdoor, rental/event, large viewing distances.

MiP: P0.9-P2.5, best value for premium applications. Excellent contrast with easy serviceability. Ideal for lobby, retail, showroom.

COB: P0.4-P1.5, highest image quality. Best for control rooms, broadcast studios, luxury hospitality, boardrooms. Module-level service acceptable.

VALUE ENGINEERING TIP

MiP delivers 85-90% of COB visual quality at 30-40% lower cost. Most consultants accept MiP as equivalent to COB for viewing distances above 2m.